

RAMONA, CA, USA

WMO: 745056

Lat: 33.038N

Lon: 116.916W

Elev: 1393

StdP: 13.97

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 53120

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.6	31.3	2.5	6.6	62.4	7.7	8.6	61.4	23.8	60.5	20.3	60.9	1.2	110	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	30.8	97.3	67.6	94.2	67.3	91.3	66.6	72.3	90.9	70.8	88.8	69.4	86.7	10.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	101.6	79.0	64.3	95.0	78.6	63.1	91.1	77.4	36.7	91.0	35.4	88.9	34.3	86.7	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	15.7	13.0	DB	24.4	104.1	2.6	3.9	22.6	106.9	21.1	109.1	19.7	111.3	17.8	114.1	
			WB	22.6	75.5	2.3	4.3	21.0	78.6	19.6	81.1	18.3	83.5	16.6	86.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	51.5	51.5	54.2	56.8	61.8	67.4	73.3	74.0	71.1	63.9	56.4	50.6	
	DBStd	10.00	6.05	5.75	5.29	5.46	4.91	4.77	4.72	4.71	5.89	5.67	5.95	5.56	
	HDD50	200	54	45	20	8	0	0	0	0	0	0	12	61	
	HDD65	2349	420	380	337	253	124	23	0	0	10	89	265	448	
	CDD50	4245	100	86	149	211	364	523	721	745	632	431	204	79	
	CDD65	920	2	1	1	6	23	96	256	281	192	55	7	0	
	CDH74	14466	98	120	200	386	698	1694	3170	3540	2741	1340	418	61	
	CDH80	6692	9	15	40	123	259	717	1535	1846	1423	602	118	5	
Wind	WSAvg	4.9	4.8	5.0	4.8	5.1	5.2	5.2	5.3	5.0	4.7	4.6	4.7	4.6	
Precipitation	PrecAvg	19.0	3.8	4.6	2.8	1.5	0.4	0.2	0.1	0.2	0.2	0.9	1.2	3.1	
	PrecMax	35.5	20.1	12.8	14.5	3.6	2.9	0.9	0.5	1.5	1.9	7.9	3.0	12.2	
	PrecMin	6.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	
	PrecStd	8.2	4.8	2.9	3.5	1.1	0.7	0.3	0.1	0.3	0.4	1.7	1.0	2.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.1	82.2	84.5	90.3	93.2	99.1	99.9	100.4	100.3	96.6	88.0	79.7	
		MCWB	56.0	55.0	58.1	58.4	62.1	66.2	70.9	68.5	69.7	62.1	57.3	54.7	
	2%	DB	76.8	77.9	79.5	83.2	87.4	92.0	94.6	96.4	96.6	91.0	83.6	75.1	
		MCWB	53.4	54.0	57.8	57.7	61.4	66.4	68.9	69.0	67.4	60.9	56.1	52.1	
	5%	DB	72.8	73.1	74.9	78.3	81.7	87.9	91.3	93.1	92.4	87.3	79.2	71.1	
		MCWB	52.1	53.3	56.4	57.3	62.1	65.6	68.5	68.1	67.1	60.3	54.5	50.9	
	10%	DB	68.5	67.5	70.4	73.0	76.8	83.6	88.5	90.2	88.7	82.1	73.8	66.2	
		MCWB	50.0	51.1	55.5	56.7	60.9	64.5	68.1	68.0	65.6	59.7	52.7	49.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	60.1	63.9	64.4	68.5	70.6	73.7	74.0	73.4	70.2	61.3	59.5
			MCDB	74.5	71.4	77.5	79.1	84.8	90.3	92.5	92.6	93.2	82.6	70.2	68.0
		2%	WB	56.8	57.2	60.7	61.7	65.7	68.4	71.9	71.9	71.4	66.0	59.3	56.8
			MCDB	67.4	69.6	73.4	75.8	81.1	87.2	89.8	90.9	90.0	80.5	73.3	66.0
5%		WB	54.7	55.2	58.6	59.5	63.7	66.8	70.4	70.4	69.6	63.8	57.6	54.6	
		MCDB	64.7	67.4	70.6	72.9	78.1	84.9	87.6	89.0	86.3	79.6	72.8	63.0	
10%		WB	53.0	53.2	56.3	57.6	61.4	65.2	69.0	69.0	67.7	61.7	55.9	52.7	
		MCDB	62.5	63.3	67.7	71.0	74.5	82.5	85.5	86.9	83.9	76.1	69.1	61.0	
Mean Daily Temperature Range	5% DB	MDBR	28.6	26.4	27.2	27.0	25.8	29.4	29.8	30.8	31.1	30.2	30.2	28.3	
		MCDBR	36.8	38.8	38.3	39.8	38.1	38.2	33.3	34.6	36.0	41.1	39.6	37.8	
		MCWBR	19.1	19.8	20.0	19.3	18.4	17.0	14.0	14.7	15.3	18.5	18.7	20.1	
	5% WB	MCDBR	29.2	31.1	32.1	32.4	31.9	34.3	29.6	30.6	31.4	32.1	32.3	28.5	
		MCWBR	17.9	18.3	19.0	17.9	16.7	16.1	12.7	13.6	14.8	17.0	18.2	17.6	
Clear-Sky Solar Irradiance	taub		0.298	0.306	0.319	0.329	0.347	0.349	0.377	0.380	0.345	0.352	0.306	0.309	
	taud		2.559	2.553	2.500	2.482	2.453	2.452	2.451	2.421	2.502	2.441	2.568	2.558	
	Ebn at Noon		294	303	305	304	298	295	286	285	292	281	289	284	
	Edh at Noon		26	29	32	35	36	36	36	36	32	31	25	24	
All-Sky Solar Radiation	RadAvg		1054	1329	1771	2147	2324	2467	2345	2226	1924	1499	1157	954	
	RadStd		68	119	93	98	141	126	128	87	84	97	63	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+1.19	N/A	N/A	+1.64	N/A	N/A	N/A	-331	+421	N/A
Regional (41 neighbors)		+1.19	N/A	-2.90	+1.35	N/A	N/A	N/A	-193	+423	+223

Nomenclature: See separate page